



Swinburne University of Technology Hawthorn Campus Department of Computing Technologies

COS20028 Big Data Architecture and Application Lab Task 4 - Semester 2

Assessment Title: Build a restaurant ordering system with Apache Kafka.

Assessment Outcome: Pass/Fail (for practice only)

Assessable Item:

- One (1) piece of a report containing the complete Kafka codes for building the restaurant order system with the screenshot of submitted commands and execution results.

Purpose of Assignment

This assignment evaluates whether a student understands how to process the streaming data via Apache Kafka.

Introduction

The project is located in the virtual machine image at the following path: **/home/training/kafka_lab/ordering_system**. For this assignment, we will simulate an ordering system within a restaurant. Our goal is using Apache Kafka to publish (random generated) orders taken from **driver thru** and **cashier** to topic(s) and, subscribe to the topics containing messages of interest to specific groups. For this assignment, we will utilize kafka-python, a Python package for Apache Kafka, to publish and subscribe to topics.

Assignment Goal

This assignment is designed to introduce students to Apache Kafka and help them develop a basic understanding of real-time data publishing and subscription.

Assignment Task

The following 2 python scripts have been provided:

1. `drive_thru.py`: Publishes orders taken from drive thru to topic(s).
2. `kitchen.py`: Subscribes to topic(s) and display only food messages on the screen using std output (ie. `print()`).

The following 2 python scripts should be accomplished:

1. `cashier.py`: Publishes orders taken from cashier to topic(s).
2. `cafe.py`: Subscribes to topic(s) and display only drink messages on the screen using std output (ie. `print()`).

The orders from drive_thru and cashier are randomly generated by OrderGenerator, which has been implemented in the virtual machine. You may examine it, but do not modify it.

After accomplishing the above 2 python scripts, execute them with "python3 cafe.py", "python3 kitchen.py", "python3 cashier.py" and "python3 drive_thru.py" in different terminals.

Note: You need to have your zoo keeper and Kafka broker up and running before executing your python scripts.

Please use the template to prepare your submission.

Submission Requirement

Only the submission using the given template will be evaluated by the tutors.

Failure to adhere to the submission requirements will immediately result in a failure for this assignment.

Rubric: Lab Task 4

Task	Assessment Outcome
Successfully finding the answer to the given question in the template and the submitted code in the report is reasonable, complete, and fully functional.	Pass/Fail

----- End of Assignment -----